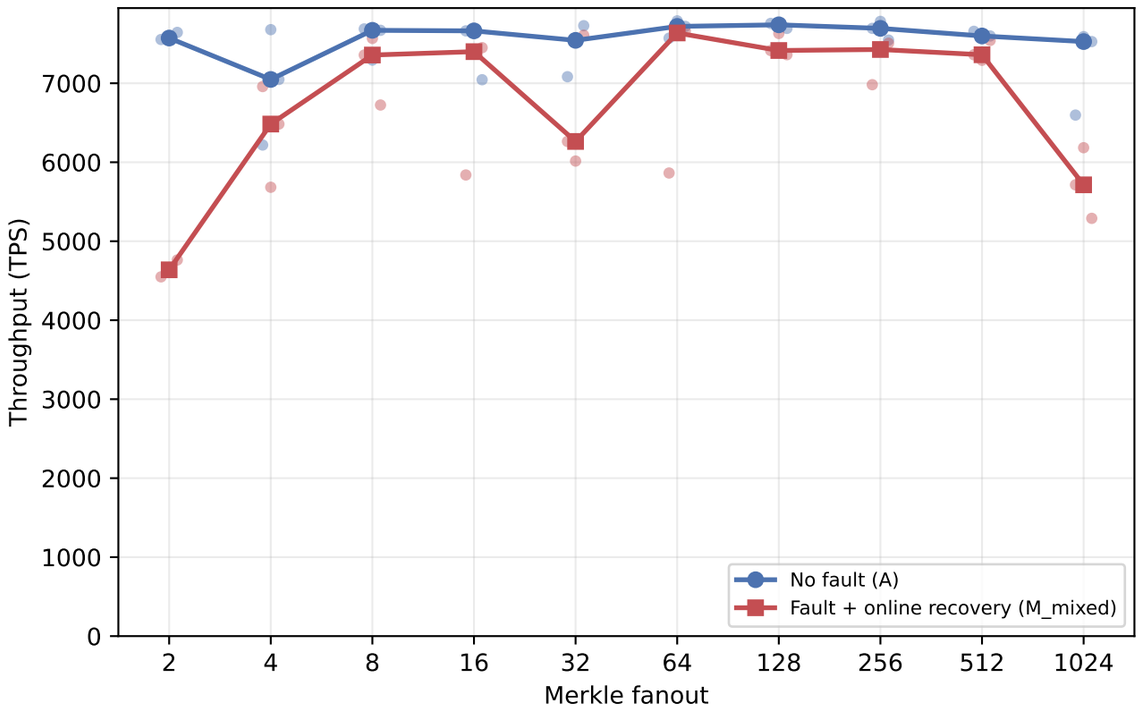
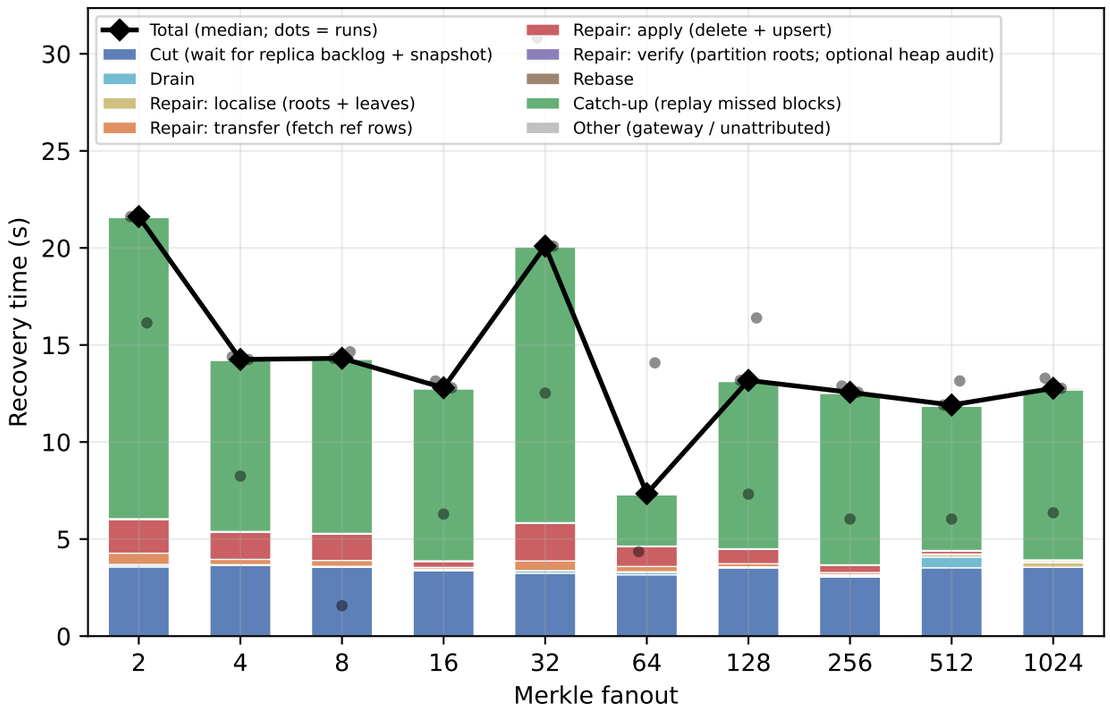
# Merkle fanout: throughput and online recovery

4,000,000 rows · P200/S1024/M256 · median of 3 runs (dots = individual runs) · complete matrix

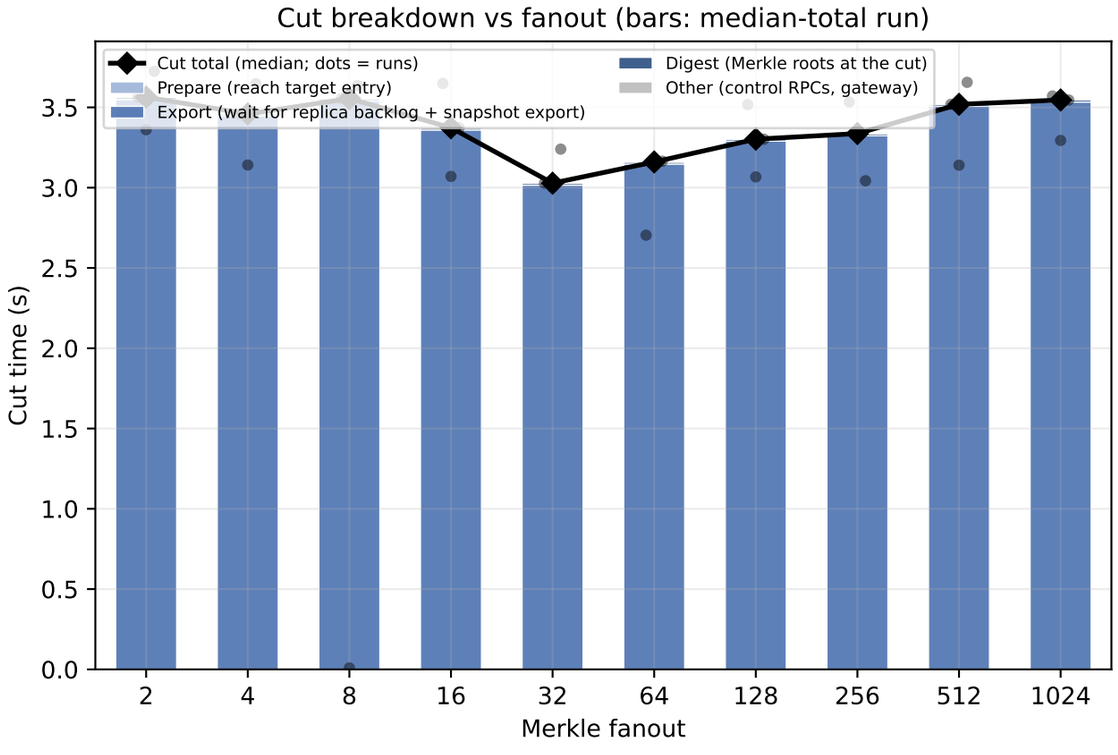
## Cluster throughput vs fanout



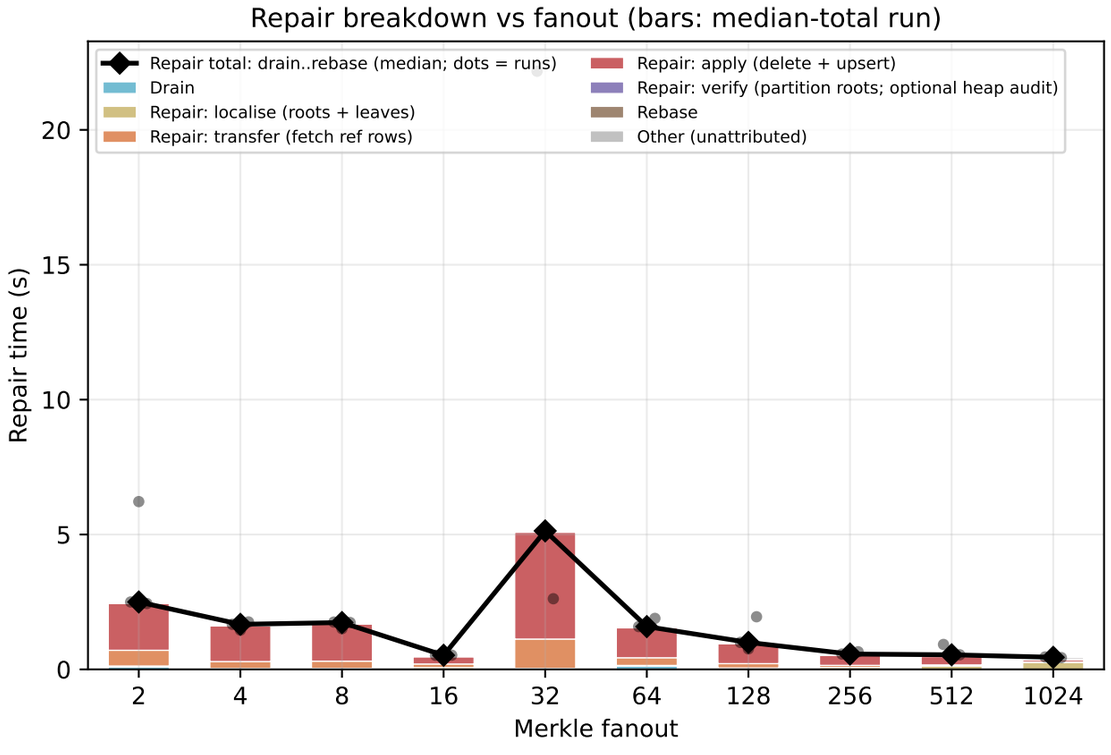

## Recovery time vs fanout — every phase (bars: median-total run)



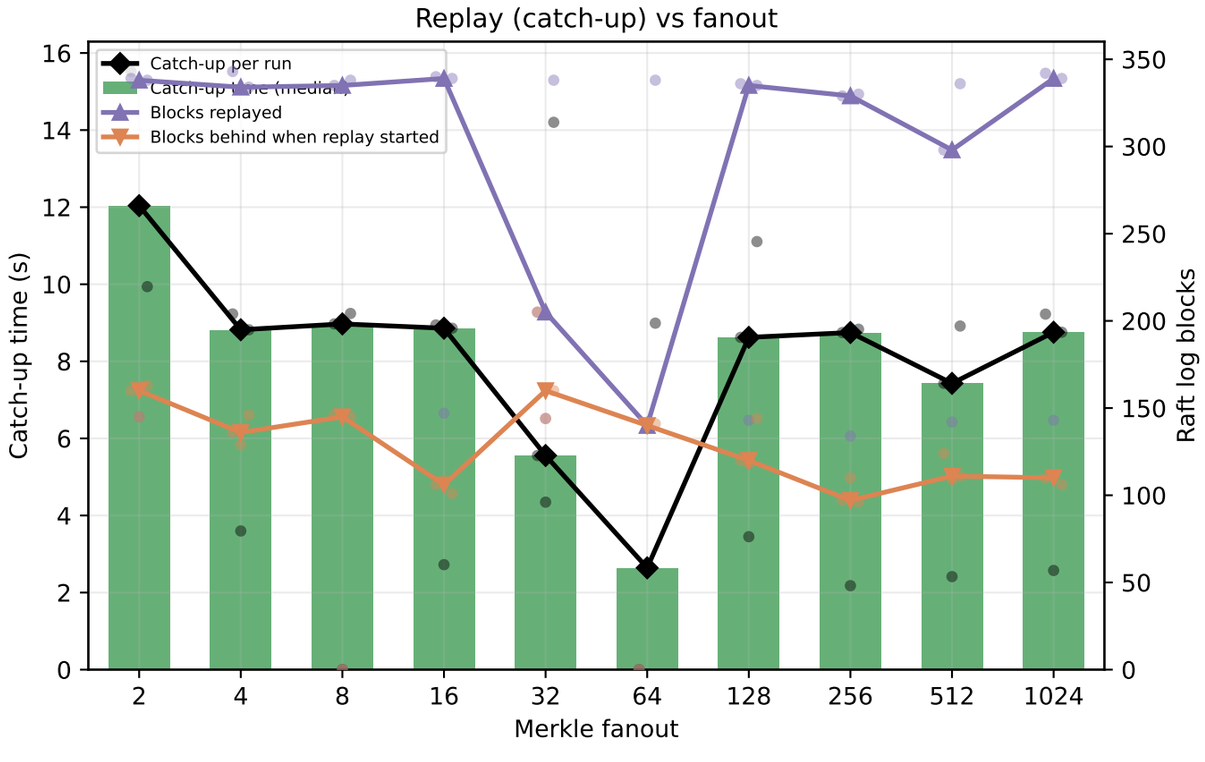

## Cut breakdown vs fanout (bars: median-total run)



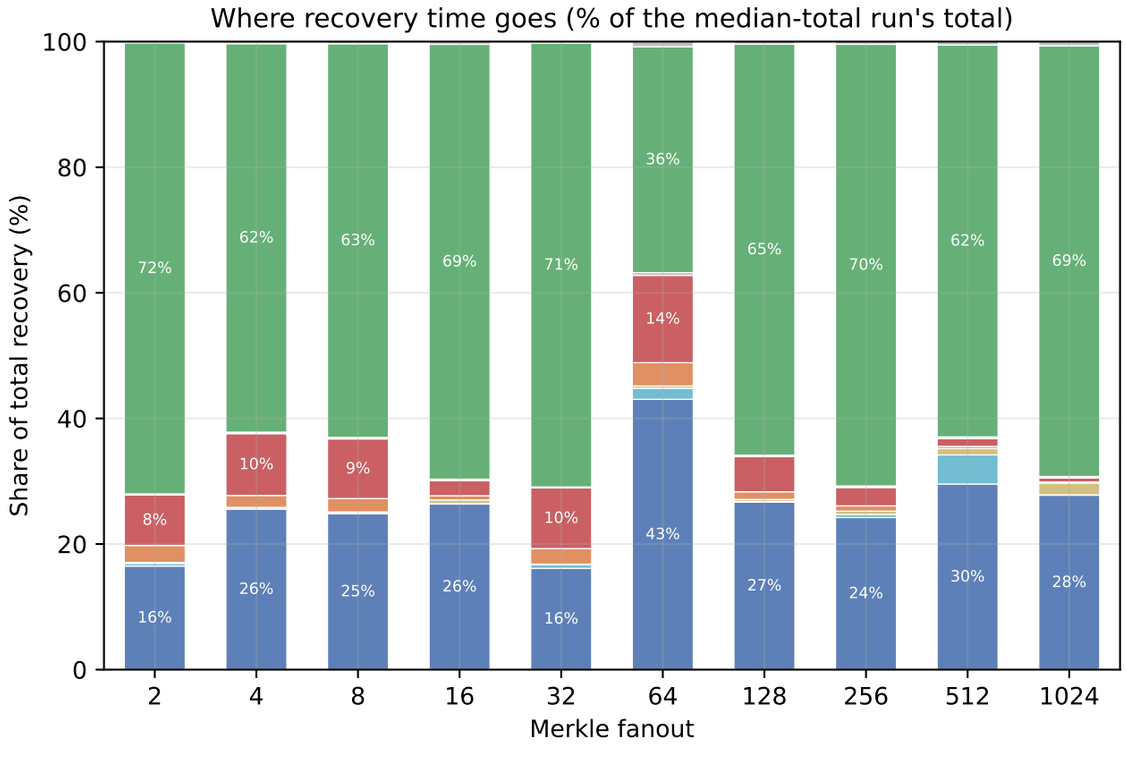

## Repair breakdown vs fanout (bars: median-total run)



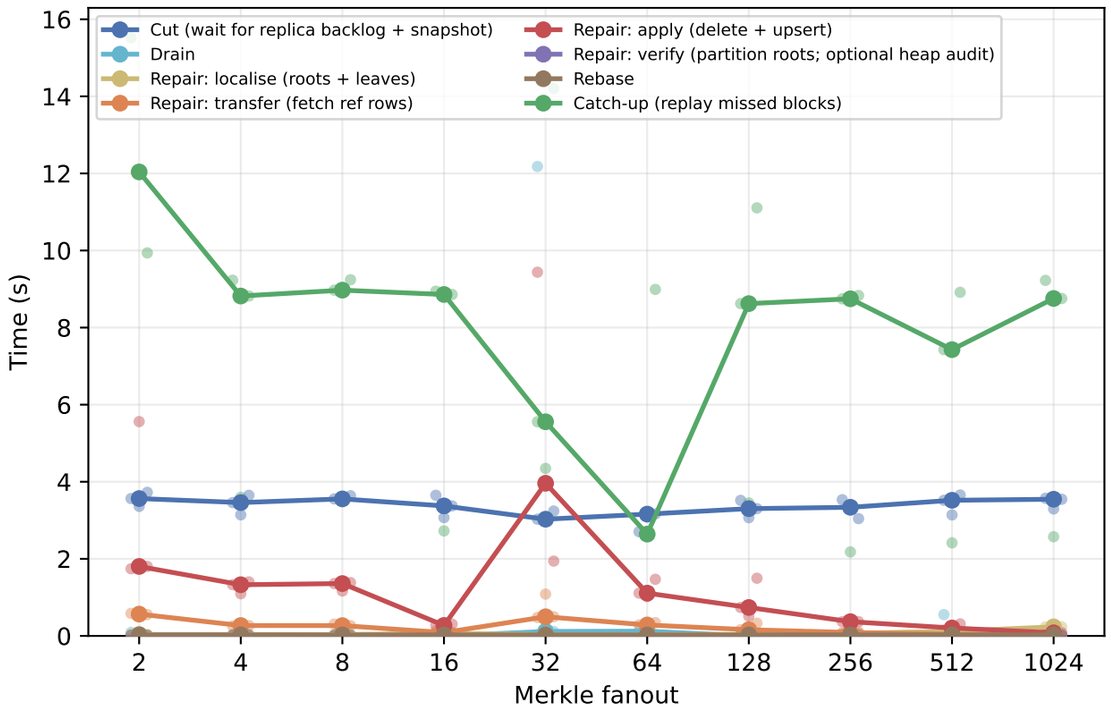

## Replay (catch-up) vs fanout




## Where recovery time goes (% of the median-total run's total)




## Each recovery phase vs fanout (median; dots = runs)




## Repair work and tree depth vs fanout

